Ultra-High Performance Liquid Chromatography coupled with Diode Array Detection and Tandem Mass Spectrometry

Ultra-High Performance Liquid Chromatography coupled with Diode Array Detection and Tandem Mass Spectrometry (UHPLC–DAD–MS/MS) analyses were performed using a Dionex Ultimate 3000 chromatographic system (Thermo Scientific) equipped with a reversed-phase C18 column (Lichrosorb RP-18, 4.6 × 200 mm, 5 µm), maintained at 30 °C. Elution was carried out in gradient mode using a binary mobile phase composed of deionized water containing 0.1% (v/v) of formic acid (solvent A) and acetonitrile (solvent B). The injection volume was 3 µL, and the flow rate was set to 0.5 mL min^{-1}. Chromatographic detection was performed at 320 nm.

The gradient elution program was as follows: 0–2 min, 25% B; 2–4 min, 25–60% B; 4–6 min, 60% B; 6–7 min, 60–95% B; 7–10 min, 95% B; 10–10.1 min, 95–25% B; and 10.1–15 min, 25% B.

Tandem mass spectrometry analysis was performed on an Ion Trap spectrometer (LCQ Fleet, Thermo Scientific) operating in positive ionization mode using full scan (m/z range 100–1000 Da). The instrument parameters were set as follows: collision-induced dissociation (CID) = 35; capillary temperature = 400 °C; sheath gas = 35 arb; auxiliary gas = 10 arb; sweep gas = 0 arb; source voltage = 5 kV; and capillary voltage = 7 V.

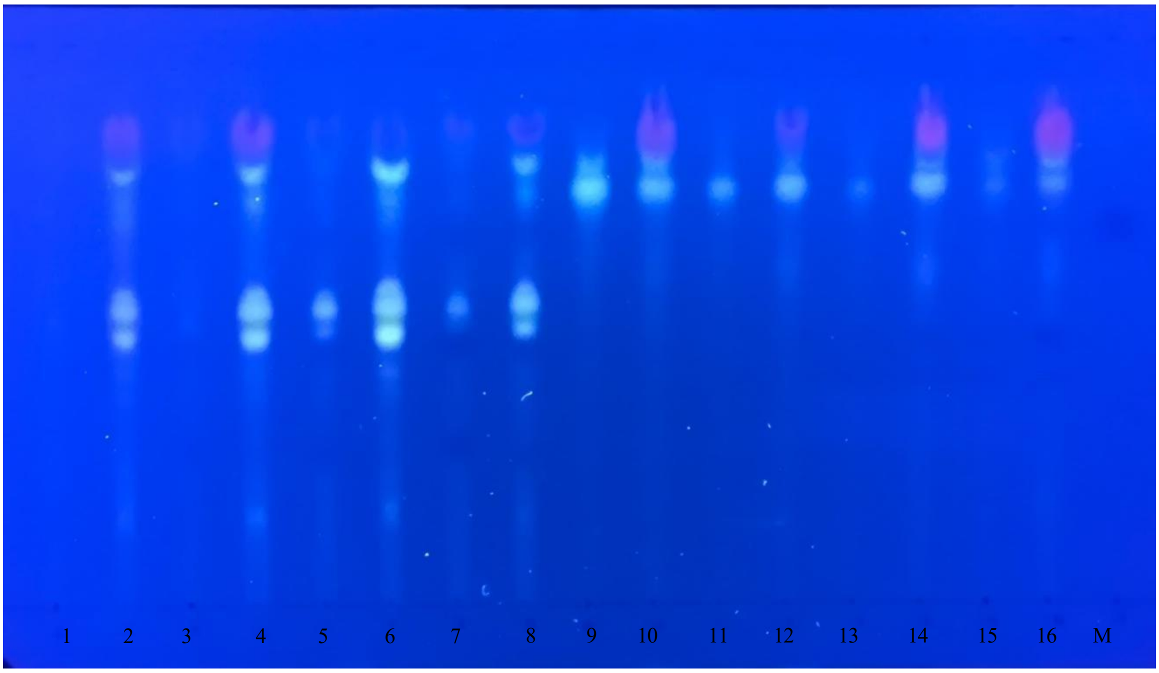
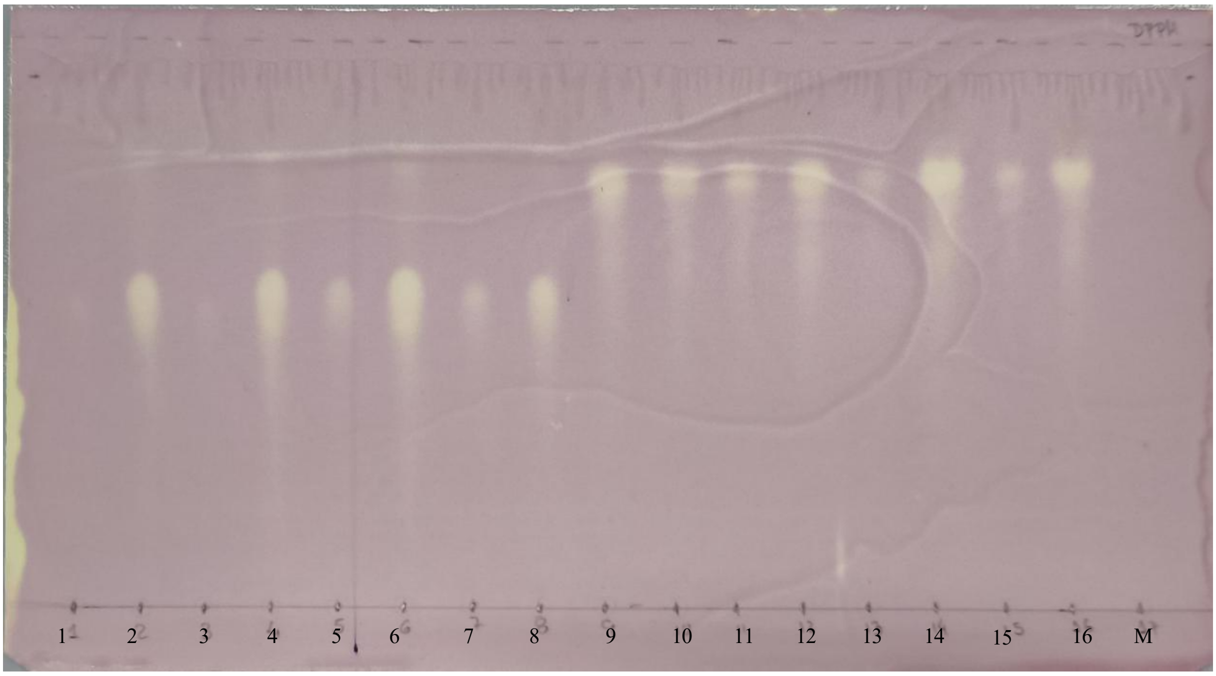


Fig. Thin-layer chromatographic profiles of the methanolic extracts from *L. esculentum* (tomato): roots (1, 3, 5, and 7 – in order: control, 250, 500, and 750 ppm) and aerial parts (stems + leaves) (2, 4, 6, and 8 – same order); *O. basilicum* (basil): roots (9, 11, 13, and 15 – in order: control, 250, 500, and 750 ppm) and aerial parts (stems + leaves) (10, 12, 14, and 16 – same order); M: reference substance (metronidazole). Stationary phase: silica gel. Mobile phase: n-butanol:acetic acid:water (4:1:1, v/v/v). Application volume: 5 µL; concentration: undetermined. Visualization: NP (1%) + UV light at 365 nm.



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